

General Navigation Test Paper

Total Questions – 200

Time allowed – 02:30 Hrs

- 1) Assuming mid-latitudes (40° to 50° N/S). At which time of year is the relationship between the length of day and night, as well as the rate of change of declination of the sun, changing at the greatest rate?
 - a. Winter solstice and autumn equinox.
 - b. Summer solstice and spring equinox.
 - c. Summer solstice and winter solstice.
 - d. Spring equinox and autumn equinox

- 2) What is the highest latitude listed below at which the sun will reach an altitude of 90° above the horizon at some time during the year?
 - a. 0°
 - b. 66°
 - c. 23°
 - d. 45°

- 3) At what approximate date is the earth furthest from the sun (aphelion)?
 - a. Beginning of July.
 - b. Beginning of January.
 - c. End of September.
 - d. End of December.

- 4) Seasons are due to the:
 - a. Earths elliptical orbit around the Sun.
 - b. variable distance between Earth and Sun.
 - c. Earths rotation on its polar axis.
 - d. inclination of the polar axis with the ecliptic plane.

- 5) What is the approximate date of perihelion, when the Earth is nearest to the Sun?
 - a. End of March.
 - b. Beginning of January.
 - c. Beginning of July.
 - d. End of December.

- 6) Sun rise at 50° N 072° E is at 0254 on 25th January. What time will the sun rise at 50° N 007° E on that day?
 - a. 2154 UTC
 - b. 0514 UTC
 - c. 0714 UTC
 - d. 0254 UTC

- 7) The compression factor of the earth:
 - a. is so small that it may be ignored when making ordinary maps and charts.
 - b. makes the difference between the polar diameter and the equatorial diameter about 22 NM.
 - c. is about 1:300.
 - d. all 3 answers are correct.

- 8) The angle between the true great-circle track and the true rhumb-line track joining the following points: A (60° S 165° W) B (60° S 177° E), at the place of departure A, is:
- 5.2°
 - 9°
 - 15.6°
 - 7.8°
- 9) An aircraft departing A ($N40^{\circ} 00' E080^{\circ} 00'$) flies a constant true track of 270° at a ground speed of 120 kt. What are the coordinates of the position reached in 6 HR?
- $N40^{\circ} 00' E068^{\circ} 10'$.
 - $N40^{\circ} 00' E064^{\circ} 20'$.
 - $N40^{\circ} 00' E060^{\circ} 00'$.
 - $N40^{\circ} 00' E070^{\circ} 30'$.
- 10) The total length of the 30° Latitude is:
- 18.706 km
 - 18.706 nm
 - 10.800 km
 - 10.800 nm
- 11) The maximum difference between geocentric and geodetic latitude occurs at about:
- 90° North and South.
 - 60° North and South.
 - 45° North and South.
 - 0° North and South (equator).
- 12) Parallels of latitude, except the equator are:
- Rhumb lines.
 - Great circles.
 - are neither Rhumb lines nor Great circles.
 - both Rhumb lines and Great circles.
- 13) A great circle track joins position A (59° S 141° W) and B (61° S 148° W). What is the difference between the great circle track at A and B?
- It increases by 3° .
 - It decreases by 3° .
 - It increases by 6° .
 - It decreases by 6° .
- 14) If an aeroplane was to circle around the Earth following parallel 60° N at a ground speed of 480 kt. In order to circle around the Earth along the equator in the same amount of time, it should fly at a ground speed of:
- 240 kt.
 - 480 kt.
 - 550 kt.
 - 960 kt.

- 15) In which two months of the year is the difference between the transit of the Apparent Sun and Mean Sun across the Greenwich Meridian the greatest?
- April and August.
 - March and September.
 - June and December.
 - February and November.
- 16) The diameter of the Earth is approximately:
- 40 000 km.
 - 18 500 km.
 - 12 700 km.
 - 6 350 km.
- 17) The circumference of the parallel of latitude at 60° N is approximately:
- 10 800 NM.
 - 18 706 NM.
 - 20 000 NM.
 - 34 641 NM.
- 18) Given: value for the ellipticity of the Earth is $1/297$. Earths semi-major axis, as measured at the equator, equals 6378.4 km. What is the semi-minor axis (km) of the earth at the axis of the Poles?
- 6 367.0
 - 6 399.9
 - 6 356.9
 - 6 378.4
- 19) The angle between the plane of the Equator and the plane of the Ecliptic is:
- 25.3 deg.
 - 65.6 deg.
 - 66.5 deg.
 - 23.5 deg.
- 20) At what latitude does the maximum difference between geodetic and geocentric latitude occur?
- 45 deg.
 - 0 deg.
 - 60 deg.
 - 90 deg.
- 21) At what approximate latitude is the length of one minute of arc along a meridian equal to one NM (1852 m) correct?
- 0°
 - 30°
 - 90°
 - 45°

- 22) An aircraft is flying around the Earth eastwards along the 60N parallel of latitude at a groundspeed of 240 knots. At what groundspeed would another aircraft have to fly eastwards along the Equator to fly once round the Earth in the same journey time?
- 240 knots.
 - 480 knots.
 - 120 knots.
 - 600 knots.
- 23) Given: The coordinates of the heliport at Issy les Moulineaux are: N48° 50 E002° 16.5 The coordinates of the antipodes are:
- S41° 10 E177° 43.5.
 - S48° 50 E177° 43.5.
 - S41° 10 W177° 43.5.
 - S48° 50 W177° 43.5.
- 24) Which of the following statements concerning the earths magnetic field is completely correct?
- At the earths magnetic equator, the inclination varies depending on whether the geographic equator is north or south of the magnetic equator.
 - Dip is the angle between total magnetic field and vertical field component.
 - The blue pole of the earths magnetic field is situated in North Canada.
 - The earths magnetic field can be classified as transient semi-permanent or permanent.
- 25) An aircraft flies the following rhumb line tracks and distances from position 04° 00N 030° 00W: 600 NM South, then 600NM East, then 600 NM North, then 600 NM West. The final position of the aircraft is:
- 04° 00N 029° 58W
 - 03° 58N 030° 02W
 - 04° 00N 030° 02W
 - 04° 00N 030° 00W
- 26) Given: Great circle from P to Q measured at P=095° Southern hemisphere Conversion angle P - Q =7° What is the rhumb line track P - Q?
- 081
 - 102
 - 088
 - 109
- 27) An RMI indicates aircraft heading. To convert the RMI bearings of NDB' s and VOR' s to true bearings the correct combination for the application of magnetic variation is:
- NDB: beacon position VOR: aircraft position
 - NDB: beacon position VOR: beacon position
 - NDB: aircraft position VOR: beacon position
 - NDB: aircraft position VOR: aircraft position
- 28) What does 5 hours 50 minutes and 20 seconds change of longitude represent?
- 81° 25
 - 35° 15

- c. $80^{\circ} 05$
- d. $87^{\circ} 35$

29) The great circle track X - Y measured at x is 319° , and Y 325° Consider the following statements:

- a. Northern hemisphere, Rhumb line track is 322° .
- b. Southern hemisphere, Rhumb line track is 322° .
- c. Southern hemisphere, Rhumb line track is 331° .
- d. Northern hemisphere, Rhumb line track is 313° .

30) On the 27th of February, at 52° S and 040° E, the sunrise is at 0243 UTC. On the same day, at 52° S and 035° W, the sunrise is at:

- a. 0243 UTC.
- b. 0523 UTC.
- c. 2143 UTC.
- d. 0743 UTC.

31) What is the definition of EAT?

- a. Estimated initial approach fix time.
- b. Estimated final approach fix time.
- c. Estimated on-blocks arrival time.
- d. Estimated time overhead the destination airfield.

32) What is the local mean time, position $65^{\circ} 25N$ $123^{\circ} 45W$ at 2200 UTC?

- a. 1345.
- b. 0615.
- c. 0815.
- d. 2200.

33) What is the meaning of the term standard time?

- a. It is another term for UTC.
- b. It is the time set by the legal authorities for a country or part of a country.
- c. It is the time zone system applicable only in the USA.
- d. It is an expression for local mean time.

34) Civil Twilight occurs between:

- a. sunset and 6 deg below the horizon.
- b. 6 deg and 12 deg below the horizon.
- c. sunrise and sunset.
- d. 12 deg and 18 deg below the horizon.

35) The main reason that day and night, throughout the year, have different duration, is due to the:

- a. inclination of the ecliptic to the equator.
- b. gravitational effect of the sun and moon on the speed of rotation of the earth.
- c. earths rotation.
- d. relative speed of the sun along the ecliptic.

36) On the earth's surface, points of zero dip are known as:

- a. isogonals.
- b. isoclinals.
- c. acclinic.
- d. agonic.

37) Required track 300° T, drift 8° right, variation 10° W, deviation -4° . What is the required compass heading?

- a. 310°
- b. 298°
- c. 306°
- d. 314°

38) The lines on the earth's surface that join points of equal magnetic variation are called:

- a. isoclines.
- b. isogrives.
- c. isotachs.
- d. isogonals.

39) The Great Circle bearing of B (70° S 060° E), from A (70° S 030° W), is approximately:

- a. 135° (T)
- b. 090° (T).
- c. 315° (T).
- d. 150° (T).

40) Isogrives are lines that connect positions that have:

- a. the same declination.
- b. the same horizontal magnetic field strength.
- c. the same variation.
- d. 0° magnetic dip.

41) Kilometre is defined as:

- a. 0,621 Statute mile.
- b. 0,454 Nautical mile.
- c. a 1/10000 part of the meridian length from Equator to the pole.
- d. the mean length of a 1/40000 part of the Equator.

42) A Nautical mile is defined as:

- a. The length of a 1 minute arc, measured anywhere on the surface of the Earth.
- b. The average length of a 1' arc of longitude and a 1' arc of latitude.
- c. 1855 metres.
- d. The average length of a 1 minute arc of a meridian

43) An isogonal is:

- a. a line of equal wind speed.
- b. a line of equal magnetic variation.
- c. a line of equal magnetic deviation.
- d. a line of zero magnetic variation.

44) An agonic line joins places of

- a. zero magnetic variation.
- b. equal magnetic deviation.
- c. equal horizontal force H.
- d. equal magnetic variation.

45) At the magnetic equator:

- a. the isogonal is an agonic line.
- b. deviation is zero.
- c. dip is zero.
- d. variation is zero.

46) An aircraft is over position HO (55° 30N 060° 15W), where YYR VOR (53° 30N 060° 15W) can be received. The magnetic variation is 31° W at HO and 28° W at YYR. What is the radial from YYR?

- a. 208°
- b. 332°
- c. 031°
- d. 028°

47) The horizontal component of the earth's magnetic field:

- a. is approximately the same at all magnetic latitudes less than 60° .
- b. weakens with increasing distance from the nearer magnetic pole.
- c. is approximately the same at magnetic latitudes 50° N and 50° S.
- d. weakens with increasing distance from the magnetic poles.

48) When accelerating on a westerly heading in the northern hemisphere, the compass card of a direct reading magnetic compass will turn:

- a. clockwise giving an apparent turn towards the north
- b. anti-clockwise giving an apparent turn towards the north
- c. anti-clockwise giving an apparent turn towards the south
- d. clockwise giving an apparent turn towards the south

49) At a specific location, the value of magnetic variation:

- a. depends on the true heading.
- b. depends on the type of compass installed.
- c. depends on the magnetic heading.
- d. varies slowly over time.

50) Where is a compass most effective?

- a. About midway between the earth's magnetic poles.
- b. In the region of the magnetic South pole.
- c. In the region of the magnetic North pole.
- d. On the geographic equator.

51) Isogonals converge at the:

- a. North magnetic pole only.

- b. North and South magnetic poles only.
- c. Magnetic equator.
- d. North and South geographic and magnetic poles.

52) The Earth can be considered as being a magnet with the:

- a. red pole near the north pole of the earth and the direction of the magnetic force pointing straight up from the earth's surface.
- b. red pole near the north pole of the earth and the direction of the magnetic force pointing straight down to the earth's surface.
- c. blue pole near the north pole of the earth and the direction of the magnetic force pointing straight up from the earth's surface.
- d. blue pole near the north pole of the earth and the direction of the magnetic force pointing straight down to the earth's surface.

53) Deviation on MH 180 is -5 and on MH 000 it is +3. Calculate coefficient C:

- a. Coefficient C = -1
- b. Coefficient C = +4
- c. Coefficient C = -2
- d. Coefficient C = +8.

54) Permanent magnetism in aircraft arises chiefly from: the combined effect of aircraft electrical equipment and the earth's magnetic field.

- a. hammering, and the effect of the earth's magnetic field, whilst under construction.
- b. the effect of internal wiring and exposure to electrical storms.
- c. exposure to the earth's magnetic field during normal operation

55) What is an advantage of a gyro-magnetic compass over a direct reading compass?

- a. Turning and acceleration errors are as large but more predictable.
- b. There is a significant reduction in lag.
- c. There is a significant reduction in turning and acceleration errors.
- d. The gyro will always indicate the vertical.

56) In a typical remote reading compass, the gyro is kept aligned with the magnetic meridian by means of:

- a. a torque motor
- b. heading selector.
- c. the Annunciator.
- d. the gyro rigidity controller.

57) The direct reading magnetic compass is made aperiodic (dead beat) by:

- a. using the lowest acceptable viscosity compass liquid.
- b. keeping the magnetic assembly mass close to the compass point and by using damping wires.
- c. pendulous suspension of the magnetic assembly.
- d. using long magnets.

58) Which one of the following is an advantage of a remote reading compass as compared with a standby compass?

- a. It is more reliable because it is operated electrically and power is always available from sources within the aircraft.
- b. It is lighter than a direct reading compass because it employs, apart from the detector unit, existing aircraft equipment.
- c. It senses the magnetic meridian instead of seeking it, increasing compass sensitivity.
- d. It eliminates the effect of turning and acceleration errors by pendulously suspending the detector unit.

59) Which of the following is an occasion for carrying out a compass swing on a Direct Reading Compass?

- a. After an aircraft has passed through a severe electrical storm, or has been struck by lightning.
- b. After any of the aircraft radio equipment has been changed due to unserviceability.
- c. Before an aircraft goes on any flight that involves a large change of magnetic latitude.
- d. Whenever an aircraft carries a large freight load regardless of its content.

60) The main reason for mounting the detector unit of a remote reading compass in the wingtip of an aeroplane is:

- a. by having detector units on both wingtips, to cancel out the deviation effects caused by the aircraft structure.
- b. to ensure that the unit is in the most accessible position on the aircraft for ease of maintenance.
- c. to minimise the amount of deviation caused by aircraft magnetism and electrical circuits.
- d. to maximise the units exposure to the earth's magnetic field.

61) On a Transverse Mercator chart, scale is exactly correct along the:

- a. prime meridian and the equator.
- b. meridian of tangency.
- c. Equator, parallel of origin and prime vertical.
- d. datum meridian and meridian perpendicular to it.

62) Scale on a Lamberts conformal chart is:

- a. constant along a parallel of latitude.
- b. constant along a meridian of longitude.
- c. varies with latitude and longitude.
- d. constant over the whole chart.

63) Transverse Mercator projections are used for:

- a. plotting charts in equatorial areas.
- b. radio navigation charts in equatorial areas.
- c. maps of large east/west extent in equatorial areas.
- d. maps of large north/south extent.

64) What is the most common use for an oblique Mercator chart?

- a. Maps of equatorial areas only.
- b. Selected great circle routes.
- c. Polar routes.

- d. Topographical maps of the British Isles.

65) The chart that is generally used for navigation in polar areas is based on a:

- a. Stereographical projection.
- b. Lambert conformal projection.
- c. Direct Mercator projection.
- d. Gnomonic projection.

66) On a Lambert Conformal Conic chart earth convergency is most accurately represented at:

- a. parallel of origin.
- b. north and south limits of the chart.
- c. Equator.
- d. standard parallels.

67) The two standard parallels of a conical Lambert projection are at N10° 40 and N41° 20. The cone constant of this chart is approximately:

- a. 0.66.
- b. 0.44.
- c. 0.18.
- d. 0.90.

68) The most likely use for an Oblique Mercator chart is:

- a. trans-polar navigation.
- b. charts for intercontinental flights following great circle tracks.
- c. global depiction of magnetic variation.
- d. maps of countries with considerable North/South extent but very little East/West extent.

69) On a Lamberts chart a straight line track crosses a meridian with a direction of 043° T and a second meridian with a direction of 055° T. If the constant of the cone is 0.75 the difference in longitude between the meridians is:

- a. 12°
- b. 9°
- c. 16°
- d. 6°

70) The standard parallels of a Lamberts conical orthomorphic projection are 07° 40N and 38° 20N. The constant of the cone for this chart is:

- a. 0.39
- b. 0.92
- c. 0.42
- d. 0.60

71) On a Lambert conformal conic chart the distance between parallels of latitude spaced the same number of degrees apart:

- a. expands between, and reduces outside, the standard parallels.
- b. is constant throughout the chart.
- c. reduces between, and expands outside, the standard parallels.
- d. is constant between, and expands outside, the standard parallels.

- 72) You are flying from A to B in the northern hemisphere. Your initial great circle track is 273° T and conversion angle is 6° . The rhumb line track from A to B is:
- 267° T
 - 276° T
 - 270° T
 - 279° T
- 73) The angular difference, on a Lambert conformal conic chart, between the arrival and departure track is equal to:
- earth convergence.
 - conversion angle.
 - map convergence.
 - difference in longitude.
- 74) An aircraft is at position A ($49^\circ 10'S$ $178^\circ 22'W$). It flies due North for 2,950 NM and then due West for 382 NM to position B. Where is position B?
- $81^\circ 40'S$ $171^\circ 50'E$.
 - $00^\circ 00'S$ $171^\circ 50'E$.
 - $00^\circ 00'S$ $175^\circ 16'E$.
 - $81^\circ 40'S$ $175^\circ 16'E$.
- 75) A Lambert conformal conic chart has a constant of the cone of 0.75. The initial course of a straight line track drawn on this chart from A ($40^\circ N$ $050^\circ W$) to B is 043° (T) at A; course at B is 055° (T). What is the longitude of B?
- $41^\circ W$.
 - $36^\circ W$.
 - $34^\circ W$.
 - $38^\circ W$.
- 76) An aircraft is at $5530N$ $03613W$, where the variation is $15W$. It is tuned to a VOR located at $5330N$ $03613W$, where the variation is $12W$. What VOR radial is the aircraft on?
- 348
 - 012
 - 165
 - 015
- 77) On a Polar Stereographic chart, the initial great circle course from A $70^\circ N$ $060^\circ W$ to B $70^\circ N$ $060^\circ E$ is approximately:
- 330° (T)
 - 210° (T)
 - 150° (T)
 - 030° (T)
- 78) Given: Waypoint 1 is $60S$ $030W$, waypoint 2 is $60S$ $020W$. What will be the approximate latitude on the display unit of an inertial navigation system at longitude $025W$?
- $6011S$
 - $6006S$

- c. 6000S
- d. 5949S

- 79) Given: Magnetic heading 311° Drift angle 10° left Relative bearing of NDB 270° What is the magnetic bearing of the NDB measured from the aircraft?
- a. 180° .
 - b. 208° .
 - c. 211° .
 - d. 221°
- 80) Two positions plotted on a polar stereographic chart, A (80° N 000°) and B (70° N 102° W) are joined by a straight line whose highest latitude is reached at 035° W. At point B, the true course is:
- a. 023°
 - b. 247°
 - c. 203°
 - d. 305°
- 81) An aircraft flies a great circle track from 56° N 070° W to 62° N 110° E. The total distance travelled is:
- a. 1788 NM.
 - b. 2040 NM.
 - c. 3720 NM.
 - d. 5420 NM.
- 82) In a navigation chart a distance of 49 NM is equal to 7 cm. The scale of the chart is approximately:
- a. 1 : 1 300 000.
 - b. 1 : 700 000.
 - c. 1 : 130 000.
 - d. 1 : 7 000 000.
- 83) An aircraft at latitude 0220 N tracks 180 T for 685 kilometres. What is its latitude at the end of the flight?
- a. 0850S
 - b. 0350S
 - c. 0250S
 - d. 0210S
- 84) The total length of the 53° N parallel of latitude on a direct Mercator chart is 133 cm. What is the approximate scale of the chart at latitude 30° S?
- a. 1 : 30 000 000
 - b. 1 : 21 000 000
 - c. 1 : 25 000 000
 - d. 1 : 18 000 000

- 85) What is the chart distance between longitudes 179° E and 175° W on a direct Mercator chart with a scale of 1: 5 000 000 at the equator?**
- 133 mm
 - 167 mm
 - 72 mm
 - 106 mm
- 86) The rhumb line distance of the 53° N parallel of latitude is measured on a Mercator chart as 133 cm; what is the scale of the chart at 30° S?**
- 1:13 000 000
 - 1:25 000 000
 - 1:30 000 000
 - 1:17 000 000
- 87) Track = 090 (T), TAS = 460 knots, W/V = 360 (T) / 100, Variation = 10 E, Deviation = - 2. What is compass heading and groundspeed?**
- 079 470 knots
 - 068 460 knots
 - 069 450 knots
 - 070 455 knots
- 88) Given: Position A is N00° E100° , Position B is 240° (T), 200 NM from A. What is the position of B?**
- S01° 40 E101° 40
 - N01° 40 E101° 40
 - S01° 40 E097° 07
 - N01° 40 E097° 07
- 89) Given: True course A to B = 250° Distance A to B = 315 NM TAS = 450 kt. W/V = 200° /60kt. ETD A = 0650 UTC. What is the ETA at B?**
- 0716 UTC.
 - 0736 UTC.
 - 0730 UTC.
 - 0810 UTC.
- 90) An aircraft is climbing at a constant CAS in ISA conditions. What will be the effect on TAS and Mach No?**
- Both increase.
 - Both decrease.
 - TAS increases and Mach No decreases.
 - TAS decreases and Mach No increases.
- 91) An aircraft has a TAS of 300 knots and is over a stretch of water between 2 airfields 500 nm apart. If the wind component is 60 knots head, what is the distance from the first airfield to the critical point?**
- 200 nm.
 - 250 nm.
 - 280 nm.

d. 300nm.

92) Given the following: True track: 192° Magnetic variation: 7° E Drift angle: 5° left What is the magnetic heading required to maintain the given track?

- a. 204°
- b. 190°
- c. 180°

93) An aircraft passes position A (60° 00N 120° 00W) on route to position B (60° 00N 140° 30W). What is the great circle track on departure from A?

- a. 279° .
- b. 288° .
- c. 261° .
- d. 270° .

94) The rhumb line track between position A (45° 00N, 010° 00W) and position B (48° 30N, 015° 00W) is approximately:

- a. 300
- b. 330
- c. 315
- d. 345

95) What is the ISA temperature value at FL 330?

- a. -66° C
- b. -50° C
- c. -56° C
- d. -81° C

96) Given the following: Magnetic heading: 060° Magnetic variation: 8° W Drift angle: 4° right What is the true track?

- a. 064°
- b. 072°
- c. 056°
- d. 048°

97) What is the longitude of a position 6 NM to the east of 58° 42N 094° 00W?

- a. 093° 53.1W.
- b. 093° 48.5W.
- c. 093° 54.0W.
- d. 094° 12.0W.

98) Given: A is N55° 000° B is N54° E010° The average true course of the great circle is 100° . The true course of the rhumbline at point A is:

- a. 096°
- b. 104°
- c. 107°
- d. 100°

99) The equivalent of 70 m/sec is approximately:

- a. 35 kt
- b. 145 kt
- c. 136 kt
- d. 210 kt

100) Given: Course 040° (T), TAS is 120 kt, Wind speed 30 kt. Maximum drift angle will be obtained for a wind direction of:

- a. 120°
- b. 130°
- c. 145°
- d. 115°

101) Pressure Altitude is 27,000 feet, OAT = -35C, Mach No = 0.45, W/V = 270/85, Track = 200T. What is drift and groundspeed?

- a. 17R / 287 knots.
- b. 15R / 310 knots.
- c. 17L / 228 knots.
- d. 18L / 252 knots.

102) Given: IAS 120 kts, FL80, COAT +20° . The TAS is:

- a. 102 kts.
- b. 132 kts.
- c. 120 kts.
- d. 141 kts.

103) An aircraft is landing on runway 23 (QDM 227°), surface wind 180° /30 kts from ATIS; variation is 13° E. The cross wind component on landing is:

- a. 26 kts.
- b. 15 kts.
- c. 20 kts.
- d. 23 kts.

104) Fuel flow per HR is 22 US-GAL, total fuel on board is 83 IMP GAL. What is endurance?

- a. 2 HR 15 MIN
- b. 3 HR 53 MIN
- c. 3 HR 12 MIN
- d. 4 HR 32 MIN

105) Given: True track 180° Drift 8° R Compass heading 195° Deviation -2°. Calculate variation?

- a. 5° W.
- b. 21° W.
- c. 25° W.
- d. 9° W.

106) Given: IAS 120 kt, FL 80, OAT +20° C. What is the TAS?

- a. 132 kt.
- b. 102 kt.

- c. 141 kt.
- d. 120 kt.

107) For a landing on runway 23 (227° magnetic) surface W/V reported by the ATIS is 180/30kt. VAR is 13° E. Calculate the cross wind component?

- a. 15 kt
- b. 26 kt
- c. 22 kt
- d. 20 kt

108) The reported surface wind from the control tower is 240° /35 kt. Runway 30 (300°). What is cross-wind component?

- a. 24 kt
- b. 30 kt
- c. 27 kt
- d. 21 kt

109) Given: Maximum allowable crosswind component is 20 kt. Runway 06, RWY QDM 063° (M). Wind direction 100° (M) Calculate the maximum allowable windspeed?

- a. 26 kt.
- b. 25 kt.
- c. 31 kt.
- d. 33 kt.

110) Given: Compass Heading 090° Deviation 2° W. Variation 12° E TAS 160 kt. Whilst maintaining a radial 070° from a VOR station, the aircraft flies a ground distance of 14 NM in 6 MIN. What is the W/V ° (T)?

- a. 160° /50 kt.
- b. 340° /25 kt.
- c. 155° /25 kt.
- d. 340° /98 kt.

111) An aircraft is maintaining a 5.2% gradient is at 7 NM from the runway, on a flat terrain; its height is approximately:

- a. 1890 FT
- b. 3640 FT
- c. 680 FT
- d. 2210 FT

112) Your pressure altitude is FL55, the QNH is 998, and the SAT is +30C. What is Density Altitude?

- a. 8620 feet.
- b. 6980 feet.
- c. 7750 feet.
- d. 10020 feet.

113) Given: FL 350, Mach 0.80, OAT -55° C. Calculate the values for TAS and local speed of sound (LSS)?

- a. 237 kt, LSS 296 kt.
- b. 461 kt , LSS 576 kt.

- c. 490 kt, LSS 461 kt.
- d. 461 kt , LSS 296 kt.

114) A flight is to be made from A 49° S 180° E/W to B 58° S, 180° E/W. The distance in kilometres from A to B is approximately:

- a. 804
- b. 1000
- c. 1222
- d. 540

115) An aircraft was over A at 1435 hours flying direct to B. Given:

Distance A to B 2900 NM

True airspeed 470 kt

Mean wind component OUT +55 kt

Mean wind component BACK -75 kt

Safe endurance 9 HR 30 MIN

The distance from A to the Point of Safe Return (PSR) A is:

- a. 1611 NM.
- b. 1759 NM.
- c. 2844 NM.
- d. 2141 NM

116) An aircraft takes-off from an airport 2 hours before sunset. The pilot flies a track of 090° (T), W/V 130° / 20 kt, TAS 100 kt. In order to return to the point of departure before sunset, the furthest distance which may be travelled is:

- a. 115 NM
- b. 97 NM
- c. 105 NM
- d. 84 NM

117) A ground feature appears 30° to the left of the centre line of the CRT of an airborne weather radar. If the heading of the aircraft is 355° (M) and the magnetic variation is 15° East, the true bearing of the aircraft from the feature is:

- a. 220°
- b. 310°
- c. 160°
- d. 130°

118) An island appears 30° to the left of the centre line on an airborne weather radar display. What is the true bearing of the aircraft from the island if at the time of observation the aircraft was on a magnetic heading (MH) of 020° with the magnetic variation (VAR) 25° W?

- a. 205°
- b. 325°
- c. 145°
- d. 195°

- 119) An island appears 60° to the left of the centre line on an airborne weather radar display. What is the true bearing of the aircraft from the island if at the time of observation the aircraft was on a magnetic heading (MH) of 276° with the magnetic variation (VAR) 10° E?**
- 086°
 - 226°
 - 046°
 - 026°
- 120) An aircraft at FL370 is required to commence descent when 100 NM from a DME facility and to cross the station at FL120. If the mean GS during the descent is 396 kt, the minimum rate of descent required is approximately:**
- 1550 FT/MIN.
 - 2400 FT/MIN.
 - 1650 FT/MIN.
 - 1000 FT/MIN.
- 121) An aircraft at FL350 is required to commence descent when 85 NM from a VOR and to cross the VOR at FL80. The mean GS for the descent is 340 kt. What is the minimum rate of descent required?**
- 1900 FT/MIN.
 - 1600 FT/MIN.
 - 1800 FT/MIN.
 - 1700 FT/MIN.
- 122) On a 12% glide slope, your groundspeed is 540 knots. What is your rate of descent?**
- 3120 feet/min.
 - 4820 feet/min.
 - 6550 feet/min.
 - 8740 feet/min.
- 123) What is effect on the Mach number and TAS in an a/c that is climbing with constant CAS?**
- Mach number increases; TAS increases.
 - Mach number remains constant; TAS increases.
 - Mach number decreases; TAS decreases.
 - Mach number increases; TAS remains constant.
- 124) An island appears 30° to the left of the centre line on an airborne weather radar display. What is the true bearing of the aircraft from the island if at the time of observation the aircraft was on a magnetic heading of 276° with the magnetic variation 12° W?**
- 318°
 - 038°
 - 234°
 - 054°
- 125) An aircraft at FL310, M0.83, temperature -30° C, is required to reduce speed in order to cross a reporting point five minutes later than planned. Assuming that a zero wind component remains unchanged, when 360 NM from the reporting point Mach Number should be reduced to:**

- a. M0.80
- b. M0.78
- c. M0.76
- d. M0.74

126) What is the validity period of the permanent data base of aeronautical information stored in the FMC In the B737-400 Flight Management System?

- a. 3 calendar months.
- b. One calendar month.
- c. 28 days.
- d. 14 days.

127) IRS differs from INS in that it:

- a. has a shorter spin-up time and suffers from laser lock.
- b. does not need to correct for Coriolis and central acceleration.
- c. has a longer spin-up time and is not affected by vertical accelerations due to gravity.
- d. does not experience Schuler errors as accelerometers are strapped down and are not rotated by a V/R feedback loop.

128) The resultant of the first integration from the north/south accelerometer of an inertial navigation system (INS) in the NAV MODE is:

- a. latitude.
- b. velocity along the local meridian.
- c. groundspeed.
- d. change latitude.

129) In an IRS:

- a. accelerometers and platform are both gyro-stabilised.
- b. accelerometers and platform are both strapped down.
- c. the platform is strapped down but the accelerometers are gyro-stabilised.
- d. the accelerometers are strapped down but the platform is gyro stabilised.

130) The principle of Schuler Tuning as applied to the operation of Inertial Navigation Systems Inertial Reference Systems is applicable to:

- a. only to strapdown laser gyro systems.
- b. both gyro-stabilised platform and strapdown systems.
- c. only gyro-stabilised systems.
- d. both gyro-stabilised and laser gyro systems but only when operating in the non strapdown mode.

131) In order to maintain an accurate vertical using a pendulous system, an aircraft inertial platform incorporates a device:

- a. with damping and a period of 84.4 MIN.
- b. with damping and a period of 84.4 SEC.
- c. without damping and a period of 84.4 SEC.
- d. without damping and a period of 84.4 MIN.

132) Some inertial reference and navigation systems are known as strapdown. This means that:

- a. the gyroscopes and accelerometers become part of the units fixture to the aircraft structure.
- b. gyros and accelerometers need satellite information input to obtain a vertical reference.
- c. only the gyros ,and not the accelerometers, become part of the units fixture to the aircraft structure.
- d. gyros, and accelerometers are mounted on a stabilised platform in the aircraft.

133) What additional information is required to be input to an Inertial Navigation System (INS) in order to obtain an W/V readout?

- a. Mach Number.
- b. TAS.
- c. Altitude and OAT.
- d. IAS.

134) What measurement is used to carry out alignment of an Inertial Navigation System?

- a. Difference in magnitude of the value of gravity compared with the gravity at the last known position.
- b. Acceleration sensed by the north gyro horizontal accelerometer.
- c. Acceleration sensed by the east gyro horizontal accelerometer.
- d. Acceleration sensed by the north gyro vertical accelerometer.

135) What is the axis of gyro drift?

- a. Horizontal.
- b. Vertical and horizontal.
- c. Vertical.
- d. Longitudinal.

136) What do you call a system of gyros and accelerometers which is fixed to the aircraft?

- a. Strapdown.
- b. Gyro-stabilised platform.
- c. Laser.
- d. Fixed.

137) The alignment time, at mid-latitudes, for an Inertial Navigation System using gimballed gyros is approximately:

- a. 20 MIN.
- b. 5 MIN.
- c. 30 MIN.
- d. 10 MIN.

138) What does the sensor of an INS/IRS measure?

- a. Velocity.
- b. Horizontal Earth Rate.
- c. Acceleration.
- d. Precession.

139) Alignment of INS and IRS equipments can take place in which of the following modes?

- a. NAV and ATT.
- b. ATT and ALIGN.

- c. ALIGN and ATT
- d. NAV and ALIGN.

140) On an IRS the modes available on the MCP are:

- a. OFF - ALIGN - NAV - ATT.
- b. OFF - STBY - ALIGN - ATT.
- c. OFF - ALIGN - NAV - STBY.
- d. OFF - STBY - ALIGN - NAV.

141) During the initial alignment of an inertial navigation system (INS) the equipment:

- a. will accept a 10° error in initial latitude and initial longitude.
- b. will not accept a 10° error in initial latitude or initial longitude.
- c. will not accept a 10° error in initial latitude but will accept a 10° error in initial longitude.
- d. will accept a 10° error in initial latitude but will not accept a 10° error in initial longitude.

142) Which of the following statement is correct concerning gyro-compassing of an inertial navigation system (INS)?

- a. Gyro-compassing of an INS is not possible in flight because it can differentiate between movement induced and misalignment induced accelerations.
- b. Gyro-compassing of an INS is not possible in flight because it cannot differentiate between movement induced and misalignment induced accelerations.
- c. Gyro-compassing of an INS is possible in flight because it cannot differentiate between movement induced and misalignment induced accelerations.
- d. Gyro-compassing of an INS is possible in flight because it can differentiate between movement induced and misalignment induced accelerations.

143) The azimuth gyro of an inertial unit has a drift of 0.01° /HR. After a flight of 12 HR with a ground speed of 500 kt, the error on the aeroplane position is approximately:

- a. 1 NM.
- b. 12 NM.
- c. 60 NM.
- d. 6 NM.

144) The period of validity of an FMS database is:

- a. varies depending on the area of operational cover.
- b. 28 days.
- c. 56 days.
- d. one week.

145) An aircraft travels from point A to point B, using the autopilot connected to the aircrafts inertial system. The coordinates of A (45° S 010° W) and B (45° S 030° W) have been entered. The true course of the aircraft on its arrival at B, to the nearest degree, is:

- a. 277°
- b. 284°
- c. 263
- d. 270°

- 146) An aircraft flying at 7,500 ft is cleared to descend to be level at 1,000 ft 6 nm before reaching a beacon. If ground speed is 156 kts, and ROD is 800 ft/min, how many miles before the beacon should descent begin?**
- a. 11.1 nm
 - b. 30.2 nm
 - c. 27.1 nm
 - d. 15.0 nm
- 147) An aircraft is in the cruise at FL155 with a TAS of 260 kts. The pilot plans a 500 ft/min ROD to arrive overhead the MAN VOR at 2,000 ft QNH 1030. Assuming a constant TAS, standard temperature and no wind, what is the minimum descent from the MAN VOR to start the descent?**
- a. 120 nm
 - b. 130 nm
 - c. 140 nm
 - d. 110 nm
- 148) An aircraft is in the cruise at FL 095 with an IAS of 155 kts. The pilot intends to descend at 500 ft/min to arrive overhead the MAN VOR at 2,000 FT QNH 1030. Assuming a constant TAS, standard temperature and no wind, what is the minimum descent from the MAN VOR to start the descent?**
- a. 38 nm
 - b. 45 nm
 - c. 48 nm
 - d. 41 nm
- 149) After flying for 16 minutes at 100 kts TAS with a 20 kts tail wind, you have to return to the airfield of departure. You will arrive after:**
- a. 10 min 40 sec
 - b. 16 min
 - c. 20 min
 - d. 24 min
- 150) An aircraft is on an IFR flight. The flight is to be changed from IFR to VFR. Is this possible?**
- a. Yes, but only with permission from ATC
 - b. No, only ATC can order you to do this
 - c. No, you have to remain IFR in accordance to the filed flight plan
 - d. Yes, the pilot in command must inform ATC using the phrase "cancelling my IFR flight"
- 151) An aircraft climbs from an airfield elevation 1,500 ft QNH 1023 hPa to FL75. What height does the aircraft have to climb? (Assume 1 hPa = 30ft.)**
- a. 7,800 ft
 - b. 6,300 ft
 - c. 6,600 ft
 - d. 6,000 ft
- 152) The fuel burn is 200 kg/h with a SG of 0.8. If the SG is 0.75 the fuel burn will be:**
- a. 200 kg/hr

- b. 267 kg/hr
- c. 213 kg/hr
- d. 188 kg/hr

153) The fuel burn of an aircraft turbine engine is 220 l/h with a SG of 0.80. If the SG is 0.75 the fuel burn will be:

- a. 206 l/hr
- b. 220 l/hr
- c. 176 l/hr
- d. 235 l/hr

154) At a fuel check you have 60 US gallons (USG) of useable fuel remaining. Alternative fuel required is 12 USG. The flight time remaining is 1 hour 35 mins. What is the highest consumption rate acceptable?

- a. 37.9 gph
- b. 21.3 gph
- c. 30.3 gph
- d. 33.0 gph

155) For a planned flight the calculated fuel is as follows:

Flight time: 2h 42min; reserve fuel at any time should not be less than 30% of the remaining trip fuel; taxi fuel: 9 kg; block fuel: 136 kg. How much fuel should remain after 2 hours flight time?

- a. 23 kg trip fuel and 10 kg reserve fuel
- b. 33 kg trip fuel and 10 kg reserve fuel
- c. 25 kg trip fuel and 8 kg reserve fuel
- d. 33 kg trip fuel and no reserve fuel

156) For a planned flight the calculated fuel is as follows:

Flight time: 3h 06min; reserve fuel at any time should not be less than 30% of the remaining trip fuel; taxi fuel: 8 kg; block fuel: 118 kg. How much fuel should remain after 2 hours flight time?

- a. 39 kg trip fuel and no reserve fuel
- b. 39 kg trip fuel and 12 kg reserve fuel
- c. 27 kg trip fuel and 12 kg reserve fuel
- d. 30 kg trip fuel and 9 kg reserve fuel

157) During a VFR flight at a navigational checkpoint the remaining usable fuel in tanks is 60 USG. The reserve fuel is 12 USG. According to the flight plan the remaining flight time is 1h 35min. What is the highest acceptable rate of consumption possible for the rest of the trip?

- a. 33.0 USG/hour
- b. 21.3 USG/hour
- c. 30.3 USG/hour
- d. 37.9 USG/hour

158) Given: Dry operating mass (DOM) 33,510 kg Traffic Load 7,600 kg Final reserve fuel 983 kg Alternate fuel 1,100 kg Contingency fuel 102 kg The estimated landing mass at alternate should be:

- a. 42,312 kg
- b. 42,093 kg
- c. 42,210 kg
- d. 42,195 kg

159) Given:

Dry operating mass (DOM) 33,510 kg

Trip fuel (TF) 2,040 kg

Alternate fuel 1,100 kg

Traffic Load 7,600 kg

Final reserve fuel 983 kg

Contingency fuel 5% of trip fuel

Which of the listed estimated masses is correct?

- a. Estimated landing mass at destination 43,295 kg
- b. Estimated take-off mass 43,295 kg
- c. Estimated landing mass at destination 43,193 kg
- d. Estimated take-off mass 45,233 kg

160) The still air distance in the climb is 189 NAM and time 30 minutes. What ground distance would be covered in a 30 kt headwind?

- a. 189 nm
- b. 193 nm
- c. 174 nm
- d. 203 nm

161) Minimum planned take-off fuel is 160 kg (30% total reserve is included). Assume the ground speed on this trip is constant. When half the distance has been flown, the remaining fuel is 70 kg. Is it necessary to divert to a nearby alternate?

- a. diversion to a nearby alternate is necessary because it is allowed to calculate the fuel without the reserve
- b. diversion to a nearby alternate is necessary because the remaining fuel is not sufficient
- c. diversion to a nearby alternate is not necessary because the reserve fuel has not been used completely
- d. diversion to a nearby alternate is necessary unless the captain decides to continue on his own responsibility

162) Given:

Distance from departure to destination 350 nm , Track 355°T , W/V 340/30 , TAS 140kts

What is the time and distance of the PET from the departure point?

- a) 75 min 211 nm
- b) 75 min 140 nm
- c) 114 min 211 nm
- d) 50 min 140 nm

163) Given:

CAS 190 kts , Altitude 9,000 ft, Temperature ISA -10°C, Track 350°T , W/V 320/40

Distance from departure to destination 350 nm, Endurance 3 hours. What is the distance to PET?

- a. 211 nm
- b. 170 nm
- c. 203 nm

d. 330 nm

164) Given:

Dry operating mass 33,510 kg

Load 7,600 kg

Final reserve fuel 983 kg

Alternate fuel 1,100 kg

Contingency fuel 102 kg

The estimated landing mass at the alternate should be:

- a. 42,093 kg
- b. 42,210 kg
- c. 42,312 kg
- d. 42,195 kg

165) Given:

Maximum allowable take-off mass 64,400 kg

Maximum landing mass 56,200 kg

Maximum zero fuel mass 53,000 kg

Dry operating mass 35,500 kg

Estimated load 14,500 kg

Estimated trip fuel 4,900 kg

Minimum take-off fuel 7,400 kg

What is the maximum additional load?

- a. 3,000 kg
- b. 5,600 kg
- c. 4,000 kg
- d. 7,000 kg

166) Given:

Maximum useable fuel 15,000 kg

Minimum reserve fuel 3,500 kg

TAS out 425 kts

Head wind component out 30 kts

Fuel flow out 2,150 kg/hr

TAS home 430 kts

Tail wind component home 20 kts

Fuel flow home 2,150 kg/hr

What is the distance to PSR?

- a. 1,143 nm
- b. 1,463 nm
- c. 1,125 nm
- d. 1,491 nm

167) Prior to an IFR flight when filling in the ATC flight plan the time information which should be entered in Item 16 (TOTAL EET) is the time elapsed from:

- a. take-off until reaching the IAF (Initial Approach Fix) of the destination aerodrome
- b. taxi-out prior to take-off until taxiing completion after landing
- c. taxiing until the IAF (Initial Approach Fix) of the destination aerodrome

- d. take-off until landing

168) In the ATC flight plan Item 10 "standard equipment" is considered to be:

- a. VHF, RTF, ADF, VOR, and transponder
- b. VHF, RTF, ADF, VOR, and transponder
- c. VHF, RTF, VOR, ILS, and transponder
- d. VHF, RTF, ADF, VOR, and ILS

169) An aircraft has a maximum certificated take-off mass of 137,000 kg but is operating at take-off mass 135,000 kg. In Item 9 (WAKE TURBULENCE CAT) of the ATC flight plan its wake turbulence category is:

- a. heavy "H"
- b. heavy/medium "H/M"
- c. medium "M"
- d. medium plus "M+"

170) An aircraft in the cruise has a calibrated airspeed of 150 kts, a TAS of 180 kts and an average ground speed of 210 kts. The speed box of the flight plan must be filled as:

- a. K0210
- b. N0180
- c. K0180
- d. N0150

171) In an ATC flight plan Item 15 (LEVEL) a cruising pressure altitude of 32,000 feet would be entered as:

- a. F320
- b. 32000
- c. FL320
- d. S3200

172) If the destination airport has no ICAO indicator, in Item 16 (DESTINATION AERODROME) of your ATS flight plan you write:

- a. AAAA
- b. ////
- c. ZZZZ
- d. XXXX

173) In an ATC flight plan Item 15 (LEVEL) a cruising pressure altitude of 32,000 feet would be entered as:

- a. F320
- b. 32000
- c. FL320
- d. S3200

174) In an ATC flight plan Item 15 (ROUTE) in terms of latitude and longitude a significant point at 41°35' north 4°15' east should be entered as:

- a. N04135E0415
- b. 41°35' N 04° 15'E
- c. N4135 E00415

- d. 4135N00415E

175) In an ATC flight plan Item 15 (ROUTE), in order to define a position as a bearing and distance from a VOR the group of figures should consist of:

- a. VOR ident, true bearing, distance in kilometers
- b. VOR ident, magnetic bearing, distance in nautical miles
- c. VOR ident, magnetic bearing, distance in kilometers
- d. full name of VOR, true bearing, distance in kilometres

176) In an ATS flight plan Item 9 (WAKE TURBULENCE CAT), an aircraft will be classified as "L" if its MTOM is?

- a. 27,000 kg
- b. 7,000 kg
- c. 57,000 kg
- d. 10,000 kg

177) In an ATS flight plan Item 15 (ROUTE), where either a route for which standard departure (SID) and a standard arrival (STAR) are provided:

- a. SID nor STAR should be entered
- b. Both should be entered
- c. STAR should be entered but not SID
- d. SID should be entered but not STAR

178) In order to comply with PANS-RAC during an IFR flight, deviations from flight plan particulars should be reported to ATC. Concerning TAS and time, the minimum deviations which must be reported are:

- a. TAS 5 kts and time 5 minutes
- b. TAS 3% and time 3 minutes
- c. TAS 5% and time 3 minutes
- d. TAS 10 kts and time 2 minutes

179) In the appropriate box of an ATC flight plan for endurance one must indicate the time corresponding to:

- a. the total usable fuel on board minus reserve fuel
- b. the total usable fuel on board
- c. the required fuel for the flight plus the alternate and 45 minutes
- d. the required fuel for the flight

180) In the ATS flight plan for a non-scheduled flight, which of the following letters should be entered in Item 8 (TYPE OF FLIGHT)?

- a. X
- b. G
- c. N/S
- d. N

181) Given the following flight plan information:

Trip fuel: 100 kg

Flight time: 1 hr 35 min

Taxi fuel: 3 kg

Block fuel: 181 kg. How should Item 19 (ENDURANCE) be completed on the ICAO flight plan?

- a. 0240
- b. 0252
- c. 0249
- d. 0204

182) When a pilot fills in an ATC flight plan he must indicate the wake turbulence category.

This category is a function of which mass?

- a. Maximum certified landing mass
- b. Estimated take-off mass
- c. Actual take-off mass
- d. Maximum certified take-off mass

183) Distance- 800Nm, Trk-270, W/v 270/10 TAS -340kts, FOB-2400 lbs, F/c-350 lbs . R/F-250lbs.

Find 1) DCP and TCP

2)DPNR and TPNR

184) If fuel consumption increase in a flight, the distance to CP will

- a. Increases
- b. Decreases
- c. Remains same

185) A and B are in the same hemisphere. The great circle bearing of A from B is 268°T and the great circle bearing of B from A is 092°T.

i) In which hemisphere are A and B ?.

ii) What is the rhumb line bearing of B from A ?

186) For a total dist of 1800 Nm with H/W of 30 Knots caluculated DCP is 1200 nm. In actual flying wind found to be 30 kts of T/W. the new DCP will be

- a. 600nm
- b. 650nm
- c. 1200nm
- d. D) 800nm

187) Pilot notices a wing tip bearing change of 12° in 3 minutes, if the ground speed of the aircraft is 150K the distance of the aircraft from the beacon is

- a. 37.5 NM
- b. 54.5 NM
- c. 65.5 NM
- d. 43.5 NM

188) The dist to PSR

- a. is maximum in nil wind conditions
- b. increases with increase in tailwind
- c. decreases with increase in tailwind

- d. is not affected by W/V

189) With a fuel of 10,000lbs PSR is calculated to be 880nm. Other factors remaining constant,if fuel is increased to 11,000lbs, dist to PSR will be

- a. 928nm
- b. 968nm
- c. 950nm
- d. 920nm

190) EICAS is:

- a. Engine indication and crew alerting system
- b. Electrical indication and crew alerting system
- c. Engine information and crew alerting system
- d. Engine indication and cabin alerting system

191) At 0400 an a/c at FL 370, GS 420 kts, is on the direct track to VOR 185 Nm distance. The a/c is required to cross VOR at FL 80. For a ROD 1800 ft/min at a GS of 250 kts, the latest time at which to commence descent is

- a. 0451
- b. 0434
- c. 0417
- d. 0428

192) OAT = +35° C, Pressure alt = 5000 feet. What is true alt?

- a. 4550 feet.
- b. 5320 feet.
- c. 4290 feet.
- d. 5550 feet.

193) Assume a Mercator chart. The distance between positions A and B, located on the same parallel and 10° longitude apart, is 6 cm. The scale at the parallel is 1: 9 260 000. What is the latitude of A and B?

- a. 30° N or S.
- b. 45° N or S.
- c. 0°
- d. 60° N or S.

194) The constant of the cone, on a Lambert chart where the convergence angle between longitudes 010° E and 030° W is 30° , is:

- a. 0.50.
- b. 0.64.
- c. 0.40.
- d. 0.75.

195) An aircraft in the northern hemisphere makes an accurate rate one turn to the right/starboard. If the initial heading was 330° , after 30 seconds of the turn the direct reading magnetic compass should read:

- a. more or less than 060° depending on the pendulous suspension used.

- b. less than 060° .
- c. more than 060° .
- d. 060°

196) Given: Position A 45° N, $?$ ° E. Position B 45° N, 45° 15E Distance A-B=280 NM B is to the East of A Required: longitude of position A?

- a. 38° 39E
- b. 49° 57E
- c. 40° 33E
- d. 51° 51E

197) Given: Distance A to B 2484 NM, Mean groundspeed out 420 kt, Mean groundspeed back 500 kt Safe endurance 08 HR 30 MIN. The distance from A to the Point of Safe Return (PSR) A is:

- a. 1908 NM.
- b. 1940 NM.
- c. 1630 NM.
- d. 1736 NM.

198) An aircraft was over Q at 1320 hours flying direct to R. Given: Distance Q to R 3016 NM True airspeed 480 kt Mean wind component OUT -90 kt Mean wind component BACK +75 kt. The ETA for reaching the Point of Equal Time (PET) between Q and R is:

- a. 1752
- b. 1756
- c. 1820
- d. 1742

199) An aircraft is flying with aid of an inertial navigation system (INS) connected to the autopilot. The following two

WPT 1: 60° N 030° W

WPT 2: 60° N 020° W When 025° W is passed the latitude shown on the display unit of the inertial navigation system will be:

- a. 60° 05.7
- b. 60° 11.0N
- c. 60° 00.0N
- d. 59° 49.0N

200) An IRS is programmed with waypoint 1: $N60^\circ$ 00 $W030^\circ$ 00, waypoint 2: $N60^\circ$ 00 $W020^\circ$ 00, waypoint 3: $N60^\circ$ 00 $W010^\circ$ 00. As the aircraft passes waypoint 2 its track will:

- a. increase by 9° .
- b. decrease by 9° .
- c. increase by 4.5° .
- d. not change.